Telemedicine Slot Booking & Prescription Portal - Microservices Architecture

UML Component Diagram - Set 1: Doctor Discovery & Slot Booking (with payment / notification hand-offs)

Client Tier

<<component>>
Patient Web / Mobile App
Patient actor

Edge / API Tier

IPatientAPI
REST / HTTPS (TLS 1.3)

<<component>>
API Gateway
auth, TLS 1.3, routing

Business Microservices

ISearchDoctors
search requests [REST]

IBooking
book / confirm slot [REST]

<<component>>
Doctor Search Service
UC-01 / FR-002

IAvailability
open-slot queries [API]

<<component>>
Booking Service
UC-02 / FR-003

IPayment
payment request [API]

<<component>>
Payment Service
UC-03 hand-off

Data & External Systems

IDoctorData
doctor queries [SQL]

ISlotStore
slot hold / reserve [SQL]

INotify
confirmation [queue]

IGateway
charge card [HTTPS]

<<component>>
Doctor Directory DB
profiles & specialties

<<component>>
Booking & Slot DB
slots, holds, bookings

<<component>>
Notification Service
email / SMS confirmation

<<external>>
Payment Gateway
external system

provided interface (ball)

required interface (socket)